

Phase 1 Data-State Analysis

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Part 1: Background & Methodology

Finding 1: Analysis Scope

We analyzed 52,048 real investment rows (filtered from 58,160 raw rows) spanning 385 funds, 101 asset managers, and 8,609 distinct company names across 6 reporting periods (December 2024 through March 2026). Of the 52,048 scorable rows, 51,629 are entity-resolvable holdings; the remaining 419 are unnamed aggregates that are scored for completeness but cannot be matched to a named company.

Where to find it:

- Full population details: `phase1_holdings_data_state_analysis.ipynb`, Section 3 (Source Asset Classification)
- Filtered row counts: `excluded_non_investment_rows.csv` (shows the 6,112 rows removed)
- Fund and manager counts: `data_state_by_fund.csv` and `data_state_by_manager.csv` (row counts in first section of notebook output)

Finding 2: Definition of Mandatory Fields

We confirmed with the Investment Office that 8 fields are mandatory for scoring completeness: Capital Invested, Current Cost, Unrealized Value, Realized Proceeds, Deal Status, Sector, Geographic Focus, and Public/Private Classification. This definition is the baseline for all downstream completeness metrics.

Where to find it:

- Exact field list and rationale: `phase1_holdings_data_state_analysis.ipynb`, Section code-imports-params
- Public/Private Classification derivation logic: Section 4.1 (code-public-private-derive)
- Public/Private methodology: `public_private_ticker_mismatches.csv` (467 rows showing why Type of Investment is authoritative)

Part 2: Completeness

Finding 3: Overall Completeness Score

Unconditional completeness is 94.6% (simple fill rate across all 8 mandatory fields, 416,384 cells scored). Conditional completeness is 97.2% (356,394 cells, after removing 59,990 cells where the field does not apply given deal status or asset type). The 2.5 percentage point difference comes from fields with conditional applicability — for example, "Realized Proceeds" should not be filled on Current deals, so a blank there is not a defect.

Where to find it:

- Unconditional fill rates: `data_state_by_field.csv`, `data_state_by_fund.csv`
- Conditional completeness breakdown: `phase1_holdings_data_state_analysis.ipynb`, Section 7.2 (code-conditional-completeness)
- Fund-level conditional scores: `data_state_by_fund_conditional.csv` (columns: `uncond_fill`, `cond_completeness`, `delta`)

Finding 4: Field-Level Completeness Ranking

Strongest fields: Geographic Focus 100%, Sector 99.8%, Unrealized Value 98.8%, Deal Status 98.2%. Weakest fields: Public/Private Classification 87.7%, Capital Invested 88.5%, Realized Proceeds 89.4%. Key insight: after conditional adjustment, Realized Proceeds and Public/Private both reach 100% — their apparent weakness is entirely non-applicability. Capital Invested does not move at all (88.5% → 88.5%), leaving 5,993 true gaps: it is the only mandatory field with a large, genuine deficit.

Where to find it:

- All 330 value columns ranked by fill rate: `data_state_by_field.csv`
- Detailed field-level analysis by fund: `data_state_by_manager_field.csv` (shows which manager is worst at which field)
- Visual: `quality_by_investment_type.png` (unconditional completeness by asset class)

Finding 5: Distribution by Fund

Unconditional: 107 funds fall below the 85% threshold (27.8%). After conditional adjustment: 50 funds (13.0%). Key finding: 58 funds were unfairly penalized — they appear incomplete only because their asset mix includes non-company holdings (Real Estate, Fund, etc.) for which Public/Private Classification does not apply. 49 funds fall below 85% on both scores; that is the

defensible escalation list. The 3 worst funds (conditional): Ab3ea87 (50.4%), Ae7bb35 (55.2%), A80d406 (61.9%).

Where to find it:

- Full fund rankings: data_state_by_fund_conditional.csv (sorted by cond_completeness)
- Detailed completeness by fund: phase1_holdings_data_state_analysis.ipynb, Section 7.2 (shows which funds benefit most from conditional adjustment)
- The 49-fund escalation list (below 85% on both scores): data_state_by_fund_conditional.csv, filter for rows where both uncond_fill and cond_completeness are < 0.85

Finding 6: Distribution by Manager — Who Reports Well, Who Doesn't?

89 managers with ≥ 10 rows scored (12 more fall in the low-evidence tail). Unconditional mean completeness 89.2%, median 94.8% — the gap between mean and median indicates a small number of poor performers drag down the average. Conditionally the same set scores 93.6% mean / 96.8% median. 9 managers report perfectly (100% unconditional). Worst manager: A5d1a01 (47.1%, 169 rows across 2 funds), followed by A99a662 (49.3%) and Aed3782 (60.6%). 26 managers sit below 85% unconditional; only 11 remain below 85% after conditional adjustment.

Where to find it:

- All manager rankings: data_state_by_manager_conditional.csv
- Manager field-by-field detail: data_state_by_manager_field.csv (shows which fields each manager is weakest on)
- Visual heatmap of 20 worst managers: manager_field_heatmap.png (note: gray cells indicate not-applicable fields, not defects)

Finding 7: Is It a Fund Problem or a Manager Problem?

Example 1: Manager A02306d has 4 funds scoring 100%, 91.7%, 83.3% and 66.7% conditionally — a 33.3-point spread proving this manager is capable of reporting well; the gaps are fund-specific, not a manager policy. Example 2: Manager A945638's 12 funds all score 100%, and A4a62c2's 4 funds all score 85.7% — identical scores across a book indicate a deliberate manager-level reporting policy. Strategic implication: tailor outreach differently (fund-level fixes vs. manager-level policy change).

Where to find it:

- Manager-to-fund mapping: data_state_by_manager.csv (columns: n_funds, n_rows per manager)
- Fund-level drill-down: data_state_by_fund_conditional.csv (can cross-reference with Investment Manager Allocator ID to see spread within one manager)
- Detailed analysis: phase1_holdings_data_state_analysis.ipynb, Section 7 (bymanager discussion)

Part 3: Data Quality & Consistency

Finding 8: Deal Status Is Trustworthy (99.9% Agreement)

Deal Status can be independently derived from Unrealized Value, Realized Proceeds, and Current Cost using the Investment Office's own definitions. On 51,113 rows where both reported and derived values exist, they agree 99.9% of the time. This proves the field is not arbitrary — there is logic behind it.

Where to find it:

- Deal Status derivation logic: phase1_holdings_data_state_analysis.ipynb, Section 7.1 (code-deal-status-derive)
- Agreement rate and conformance table: consistency_rule_results.csv (row: "Written Off => Unrealized = 0..." and related Deal Status rules all show 99%+ pass rates)
- Full conformance details: deal_status_conformance_exceptions.csv (the 26 exceptions, showing CSV vs. derived)

Finding 9: The 26 Deal Status Anomalies (All One Type)

All 26 disagreements follow the same pattern: CSV reports "Written Off" but the numbers suggest "Fully Exited." They concentrate in a small number of managers. These are ready-made candidates for PDF validation.

Where to find it:

- Exact list of 26 exceptions: deal_status_conformance_exceptions.csv (columns: Fund ID, Manager ID, Source Asset, As At Date, reported Deal Status, derived Deal Status, Unrealized Value, Realized Proceeds, Current Cost)
- Why they matter: these are the same 26 rows flagged by consistency_rule_results.csv (row: "Written Off definition conformance")

Finding 10: Consistency Rules — 16 Internal Checks

11 of 16 rules pass at $\geq 99\%$ (data structure is sound). 5 rules show noteworthy violations:

- **Capital Invested non-decreasing over time: 3.77% fail (1,380 rows)** — Cumulative invested amount decreased between quarters; suggests restatement or extraction error.
- **Total Value = Unrealized + Realized: 1.34% fail (624 rows)** — Accounting identity broken; may reflect rounding, currency, or reporting error (1% tolerance applied).
- **Realized Proceeds non-decreasing over time: 1.29% fail (475 rows)** — Cumulative proceeds decreased; again suggests restatement.
- **Written Off definition (UV=0, RP=0, CC=0 or blank): 1.05% fail (26 rows)** — The same 26 Deal Status anomalies.
- **Deal Status does not regress: 0.58% fail (300 rows)** — Position moved backward in the deal lifecycle (e.g., Fully Exited reverting to Current).

The other 11 rules (non-negative values, Deal Status definitions for Current/Partially Exited, date ordering, cost limits) all pass at 99%+.

Where to find it:

- Full results with applicable row counts and violation counts: `consistency_rule_results.csv` (all 16 rules listed)
- Detailed rule logic: `phase1_holdings_data_state_analysis.ipynb`, Section 12 (code-`consistency-checks`)
- CSV-to-PDF handoff: `deal_status_conformance_exceptions.csv` (for the 26 Written Off anomalies)

Part 4: Operational Insights

Finding 11: Reporting Gaps

Raw data shows 511 "missing" quarterly reports (5.0% of expected series). After adjusting for each fund's own reporting rhythm (some funds report annually, semi-annually, or only once), true gaps shrink to 144 (1.4%). Most "missing" quarters are actually correct: annual-reporting funds report once per year, semi-annual funds skip quarters. This prevents false escalation.

Where to find it:

- Reporting cadence by fund: `fund_reporting_cadence.csv` (quarterly/annual/semi-annual/single-period)

- Real gaps (only): reporting_gaps_by_series.csv (144 rows; columns: Fund ID, Source Asset, first/last date, cadence, gap_periods)
- Detailed gap analysis: phase1_holdings_data_state_analysis.ipynb, Section 13 (cadence-aware gap detection)

Finding 12: Data Quality Varies by Asset Type

Unconditional fill rate by asset type — strong performers: Public Company 97.2% (3,665 rows), Private Company 96.6% (41,985 rows). Weaker: Cryptocurrency 85.4% (535 rows), Infrastructure 83.1% (48), Fund 81.6% (3,101), Financial Asset 80.6% (218), Real Estate 72.2% (991), Natural Resources 50.0% (n=6, small sample). Explanation: much of this gap is structural — Public/Private Classification is by definition blank on every non-company asset type, which alone accounts for the Crypto, Infrastructure and Fund shortfalls. Real Estate additionally uses property-level cost and proceeds conventions rather than equity-style valuations.

Where to find it:

- Completeness by asset type (unconditional): quality_by_investment_type.png
- Note: the notebook currently outputs the unconditional version only; a conditional split by asset type has not been run
- Field-level breakdown by asset: code-investment-type output in notebook (shows which fields drive gaps for Real Estate vs. other types)
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Finding 13: Public vs. Private Classification

Type of Investment is the authoritative source (99.998% filled; only 1 row missing out of 58,160). Ticker/Exchange fields are far too sparse (Public Market Ticker 8.8%, Exchange 8.8%) to serve as a proxy. Decision: Public/Private Classification is derived directly from Type of Investment (only "Public Company" and "Private Company" receive values; all other asset types are marked not-applicable, not missing). Evidence: 467 rows carry a ticker but are not typed Public Company, confirming ticker presence is not a reliable signal.

Where to find it:

- Derivation logic: phase1_holdings_data_state_analysis.ipynb, Section 4.1 (code-public-private-derive)
- Evidence (ticker sparsity vs. Type of Investment): public_private_ticker_mismatches.csv (467 rows showing ticker/Type mismatches; analysis confirms Type is authoritative)
- Conditional applicability rule: Section 7.2, TYPE_COND logic in code-conditional-completeness

Finding 14: Entity Overlap

8,609 distinct company names in the entity-resolvable population (51,629 rows). 1,098 of them (12.8%) are held by more than one fund — this is the addressable set for matching and cross-fund valuation consistency. A further 424 groups collide after normalisation (e.g. "X.AI" vs "X.AI Holdings Corp."), and 1,271 fuzzy pairs score ≥ 88 similarity. Note on scope: Phase 3 entity resolution ran over the full extract (8,706 distinct Source Asset values across all 58,160 rows) and reduced it to 8,220 canonical names; the two counts differ because they use different denominators, not because either is wrong.

Where to find it:

- Company name inventory: holdings_attributes.csv (row counts per unique name)
- Multi-fund holdings: phase1_holdings_data_state_analysis.ipynb, Section 14 (entity resolution scope)
- Spelling variation candidates: same section (424 normalized collision groups; 1,271 fuzzy pairs at ≥ 88 similarity in entity_resolution_candidates.csv)

Part 5: Data Hygiene

Finding 15: What Was Filtered Out and Why

6,112 rows excluded from completeness scoring (10.5% of raw data):

- 5,518 Balance Sheet lines (cash, liabilities) — not investments
- 577 Accounting entries mislabeled as holdings (e.g. "Quarterly Unrealized Gain/Loss") — valuation adjustments, not holdings
- 17 Subtotal rows — double-counting prevention
- Separately, 419 unnamed aggregates ("Other Investments", "Seed Deals") are KEPT in the scorable population — they are real exposure and are scored for completeness, but they are excluded from entity resolution because they cannot be resolved to a specific company. 52,048 scorable = 51,629 holdings + 419 unnamed aggregates.

Where to find it:

- Detailed exclusion list with reasons: excluded_non_investment_rows.csv (columns: Source Asset, reason_for_exclusion, Fund ID, manager)
- Classification logic: phase1_holdings_data_state_analysis.ipynb, Section 3 (source_asset_filter.py; also Section 3.1 output showing breakdown by category)
- Source asset summary: source_asset_classification.csv (row counts by category: holding / unnamed_aggregate / balance_sheet / accounting_entry / subtotal)

Finding 16: Interesting Discovery

One accounting entry, "Quarterly Unrealized Gain/Loss," appeared as a holding row 337 times across 67 different funds — it was among the most frequently repeated "company names" in the entire dataset before filtering. Our classification rule caught and excluded it, preventing a phantom entity from polluting entity matching and from being counted as a widely-held portfolio company.

Where to find it:

- Example rows: `excluded_non_investment_rows.csv` (filter for Source Asset contains "Unrealized Gain/Loss")
- Notebook discussion: `phase1_holdings_data_state_analysis.ipynb`, Section 3 (example of why classification matters)
- Impact on entity count: had it survived the filter it would have ranked as the single most widely held "company" in the book, ahead of Stripe (15 funds).

Data Files Available for Slides & Reporting

For Charts and Rankings:

- `data_state_by_fund_conditional.csv` — all 385 funds ranked by conditional completeness
- `data_state_by_manager_conditional.csv` — all 101 managers listed (89 have ≥ 10 rows and are scored)
- `data_state_by_field.csv` — all 330 value columns ranked by fill rate (includes the 8 mandatory fields)
- `data_state_by_manager_field.csv` — manager-by-field matrix (heatmap source)
- `fund_reporting_cadence.csv` — distribution of reporting rhythms

For Specific Cases & Examples:

- `deal_status_conformance_exceptions.csv` — the 26 anomalies with side-by-side comparison
- `consistency_rule_results.csv` — all 16 rules with pass/fail counts
- `reporting_gaps_by_series.csv` — the 144 true gaps with fund and holding names

Reference & Supporting:

- `public_private_ticker_mismatches.csv` — evidence for why ticker is not authoritative
- `excluded_non_investment_rows.csv` — explains why $52,048 \neq 58,160$
- `source_asset_classification.csv` — row and name counts by filter category
- `holdings_attributes.csv` — full 837-column inventory (reference only; 249 columns are 100% empty)

Charts:

- completeness_by_dimension.png — fund vs. manager distribution side-by-side
- fund_quality_distribution_outliers.png — fund histogram with outlier tail marked
- manager_field_heatmap.png — 20 worst managers, field-by-field (gray = not applicable)
- portfolio_concentration.png — top sectors and companies by fund count
- quality_by_investment_type.png — asset type completeness (unconditional version)